

Interpretation

1 - In Figure 1 we have a data distribution, the dots represent the sparse data for the axis X and Y, and the lines represent the fit of a hypothetical classification model. Based on the distributions of Figure 1:

- Which distribution has the best balance between bias and variance?
- Describe your thoughts about your selection.

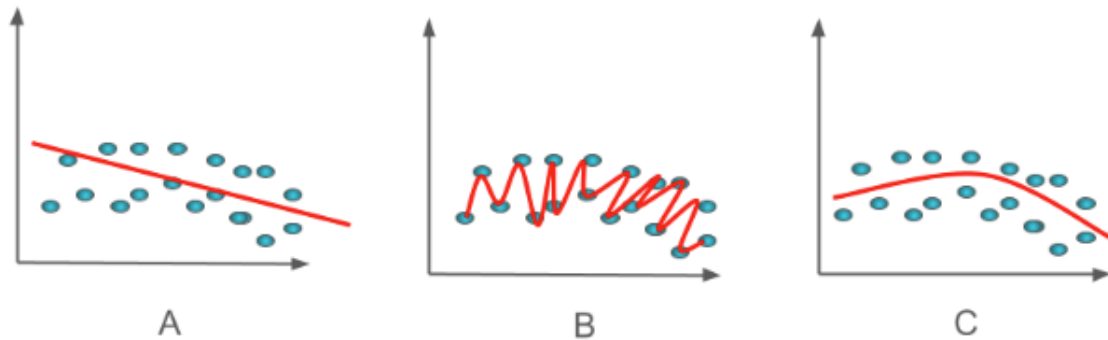


Figure 1 - Data distribution samples

Option **C** is the most appropriate model because it captures the underlying structure of the data while maintaining good generalization ability.

In option **A**, the straight line fails to capture the obvious curve in the data distribution. This is a classic example of underfitting, where the model has high bias and low variance. This issue often suggests that the model was trained with insufficient training data, number of epochs, or is structurally too simplistic for the task.

In option **B**, the model is extremely flexible and adjusts too closely to the training data. It captures not only the true pattern but also the noise, meaning it will struggle significantly to generalize and will likely perform poorly on unseen data.

2 - Figure 2 presents a simple graph with 2 curves and 1 line. In model selection and evaluation:

- What is the purpose of this graph and its name?
- What kind of model result does the dashed line represent?
- Which curve represents a better fit, the red or the green? Why?
- Describe your thoughts about your selection.

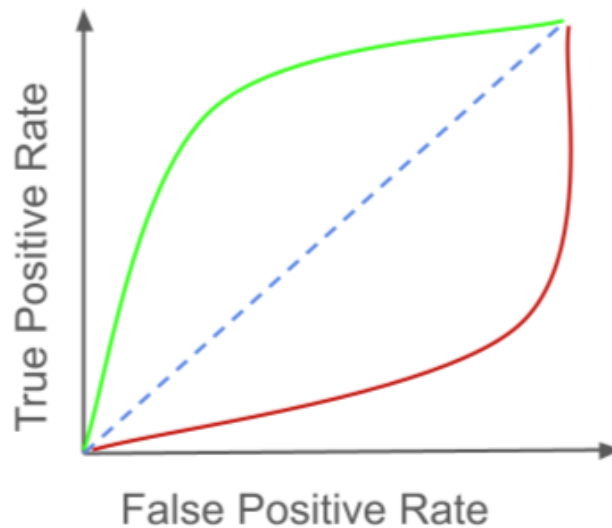


Figure 2 - Simple graph

This graph is known as a Receiver Operating Characteristic (ROC) curve. It is a fundamental tool for evaluating the performance of classification models across various classification thresholds.

The blue dashed diagonal line represents a random classifier. A model whose performance tracks along this line has an Area Under the Curve (AUC) of 0.5, meaning it has zero discriminative power and is effectively making random predictions. In other words, It performs like flipping a coin.

The ideal point on an ROC curve is the top-left corner, which corresponds to a 100% True Positive Rate and a 0% False Positive Rate. The green curve arcs closely toward this ideal top-left corner, demonstrating that it can achieve a high rate of true positive classifications before it starts accumulating false positives. Conversely, the red curve dips *below* the line of random guessing. This indicates a model with an AUC of less than 0.5, which essentially means the model is consistently predicting the wrong class. In practice, if a model consistently performs this poorly, simply inverting its predictions would yield a better result.

3 - Figure 3 presents classification model training and the evaluation. This model classifies 3 classes (A, B, C). Graph A represents the training accuracy over the epochs, Graph B represents the training loss over the epochs, and the table represents the evaluation of the model using some test samples, we used a confusion matrix to evaluate the classes trained.

- Can we say that the model has a good performance in the test evaluation?
- What phenomenon happened during the test evaluation?
- Describe your thoughts about your selection.

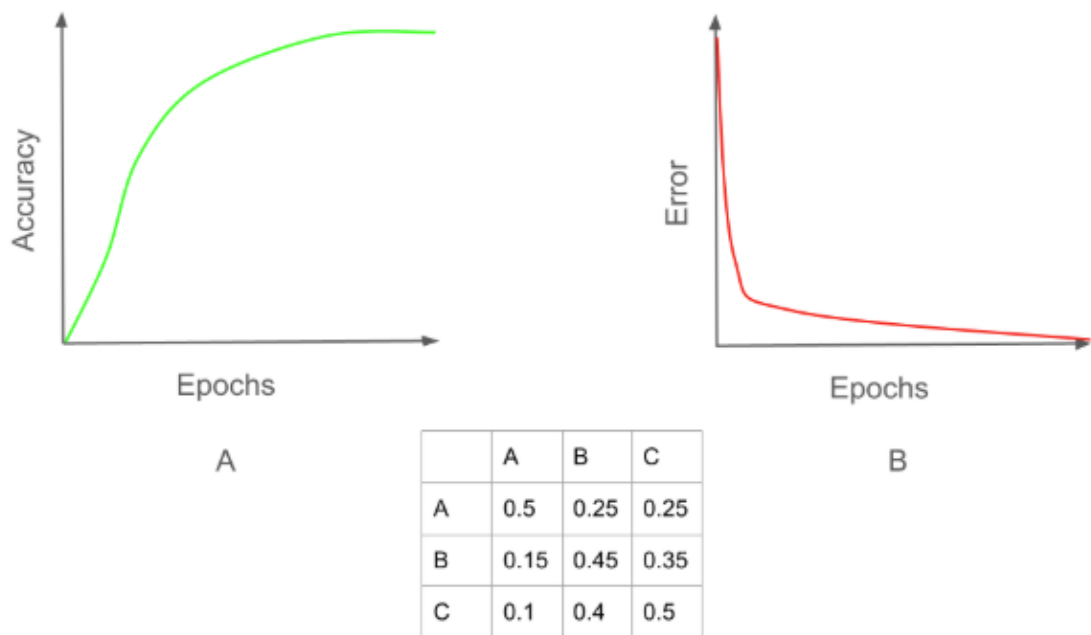


Figure 3 - Model train and evaluation pipeline

No, the model does not have good performance in the test evaluation. Looking at the main diagonal of the confusion matrix (which represents correct predictions or True Positives), the model correctly classifies the samples only 45% to 50% of the time.

The specific phenomenon happening here is overfitting. The model has essentially memorized the training data but completely failed to generalize those patterns to unseen test data.

From the training curves: Graph A shows the training accuracy rising smoothly and plateauing at what appears to be a very high level. Simultaneously, Graph B shows the training error (loss) dropping rapidly and plateauing near zero. This tells us the model learned the *training* dataset almost perfectly. However, the test confusion matrix shows that the model does not generalize well to unseen data. This mismatch between strong training performance and weaker test performance is characteristic of overfitting.